D88 | Income is not cash

Private synthetic twelve-month B case | Lakh rupees | 13 November 2026 | Build on a blank sheet first; use these authored formulas only to check your work.

Scope and changes	The D85/86 scenario is retained. D88 adds depreciation 2 to both plans, an opening cash/PPE/equity ledger and no new economic benefit. No tax, debt, inventory, payables, interest, disposals, dividends or financing.
Input -> process -> output	Inputs holds invented values; Calculations links statements; Output reports decision limits; Audit reconciles identities; Sensitivity retains no-bet/base/downside/upside; Registry records input lineage. Yellow cells are editable; purple cells calculate.
Controlled changes	Change only growth collections +1: AR -1/cash +1, NI/assets unchanged. Growth capex +1: PPE +1/cash -1, NI unchanged. Growth variable cost +1: NI/cash/equity -1. Restore each before the next test.
Validity is not solvency	Non-numeric, blank, negative or impossible inputs flag invalid and suppress outputs. Negative closing cash is not an invalid input by itself: it is a funding failure to report, not a value to hide or plug.
Decision gates	Cash floor 2 and incremental operating target 2 are stipulated. Growth ends 1.1 and fails the endpoint floor. Upside ends 5.9 but dated cash events and quality evidence are absent. Endpoint adequacy cannot prove cash never breaches the floor.
Separate valuation lab	Valuation is an independent today-price two-year real/nominal consistency example. Its NPV is not added to SME cash, income, equity or any growth benefit. All flows use the same general inflation assumption, no tax.
Execution and boundaries	Failure log records actual native test outcomes on this authored workbook. A balanced sheet can still be wrong. This is not an independently audited learner workbook, Excel/Sheets parity, real accounts, an actual external review or authority to spend/share.

Source practice: ICAEW 20 principles (input/process/output, controls, changed-case and invalid-input tests). ACCA inflation appraisal supports matching cash-flow price basis to rates; historical tax shortcuts are not adopted.

Inputs | Four coherent scenarios

All entries are invented classroom assumptions, not company observations. B is one forward-looking twelve-month period. Opening balances are at its start.

Input / lakh rupees	No bet	Growth base	Growth downside	Growth upside
Revenue	30.00	36.00	30.00	42.00
Variable cost paid	18.00	21.60	18.00	25.20
Fixed cost paid	8.00	10.00	10.00	10.00
Collections in B	27.00	25.20	21.00	33.60
Capex paid in B	0.00	2.50	2.50	2.50
Depreciation in B	2.00	2.00	2.00	2.00
Opening cash	10.00	10.00	10.00	10.00
Opening PPE	20.00	20.00	20.00	20.00
Opening AR	0.00	0.00	0.00	0.00
Opening equity	30.00	30.00	30.00	30.00

Endpoint cash floor	2.00
Incremental operating target	2.00
Quality target (unmeasured)	95.00%

Pasted blanks/text may bypass UI validation, so Calculations separately checks COUNT and feasible accounting bounds. Depreciation cannot exceed available PPE. Collections cannot exceed opening AR plus revenue. Opening ledger must balance independently.

Calculations | Link every bridge

No balancing plug: cash comes from movements; equity from opening equity plus net income; PPE from capex/depreciation. Liabilities are zero by the case exclusions.

Measure / lakh rupees	No bet	Growth base	Growth downside	Growth upside
Valid numeric/feasible inputs? (1=yes)	1.0000	1.0000	1.0000	1.0000
Revenue	30.0000	36.0000	30.0000	42.0000
Operating remainder before dep.	4.0000	4.4000	2.0000	6.8000
Depreciation	2.0000	2.0000	2.0000	2.0000
Net income	2.0000	2.4000	0.0000	4.8000
AR increase	3.0000	10.8000	9.0000	8.4000
CFO	1.0000	-6.4000	-7.0000	-1.6000
CFI	0.0000	-2.5000	-2.5000	-2.5000
CFF (excluded)	0.0000	0.0000	0.0000	0.0000
Net cash movement	1.0000	-8.9000	-9.5000	-4.1000
Closing cash	11.0000	1.1000	0.5000	5.9000
Closing AR	3.0000	10.8000	9.0000	8.4000
Closing net PPE	18.0000	20.5000	20.5000	20.5000
Closing assets	32.0000	32.4000	30.0000	34.8000
Closing equity	32.0000	32.4000	30.0000	34.8000
Operating margin	13.33%	12.22%	6.67%	16.19%

The CFO add-back reverses depreciation once. Capex affects investing cash and PPE, not operating expense. AR increase is revenue minus collections under the explicit opening/exclusion assumptions; no allowance, tax or FX model is implied.

Audit | Zero is necessary, not enough

All financial reconciliation amounts should be zero with valid inputs. Formula lineage and controlled mutations are separately tested; a copied plug can hide errors.

Check / expected zero	No bet	Growth base	Growth downside	Growth upside
Opening assets - equity	0.0000	0.0000	0.0000	0.0000
Closing assets - equity	0.0000	0.0000	0.0000	0.0000
Cash roll-forward residual	0.0000	0.0000	0.0000	0.0000
Revenue - collections - AR change	0.0000	0.0000	0.0000	0.0000
PPE roll-forward residual	0.0000	0.0000	0.0000	0.0000
Equity roll-forward residual	0.0000	0.0000	0.0000	0.0000
Inputs/reconciliation valid? (1=yes)	1.0000	1.0000	1.0000	1.0000

A broken reference, non-numeric cell or invalid ledger makes the valid/reconciled flag zero. Do not interpret the flag as accuracy of forecasts, independent review, audit assurance or real authority.

Output | Funding and evidence still matter

Baseline and growth base from referenced statements. All cash is endpoint cash, not measured minimum cash during B. No real funding or quality observation exists.

Measure / lakh rupees	No bet	Growth base	Growth - no bet
Operating remainder	4.0000	4.4000	0.4000
Net income	2.0000	2.4000	0.4000
CFO	1.0000	-6.4000	-7.4000
CFI	0.0000	-2.5000	-2.5000
Cash movement	1.0000	-8.9000	-9.9000
Ending company cash	11.0000	1.1000	-9.9000
Ending assets	32.0000	32.4000	0.4000

Endpoint growth floor status	FAIL: endpoint below floor

Growth base ends 1.1 below floor 2, with incremental operating 0.4 below 2. Do not approve an unconditional launch. Peak funding and cash-never-below-floor remain UNKNOWN without dated flows; delivery quality remains UNKNOWN without evidence.

Fictional review gates: day 0 data/formula reviewer checks all input/source IDs; day 30 collections/quality reviewer requests dated evidence; day 60 formula reviewer reruns changed-case/invalid tests; day 90 permission reviewer revisits defer/redesign. Stop on invalid formulas, unfunded floor breach or missing evidence. These roles are simulated, not actual signoffs.

Sensitivity | Same counterfactual

No-bet plan remains the comparator. Equal depreciation leaves incremental NI equal to incremental pre-depreciation operating remainder; probabilities are not assigned.

Measure / lakh rupees	Growth base	Downside	Upside
Incremental operating	0.4000	-2.0000	2.8000
Incremental net income	0.4000	-2.0000	2.8000
Incremental cash movement	-9.9000	-10.5000	-5.1000
Ending company cash	1.1000	0.5000	5.9000
Endpoint floor margin	-0.9000	-1.5000	3.9000
Cash-never-below-floor?	UNKNOWN: dates absent	UNKNOWN: dates absent	UNKNOWN: dates absent
Delivery quality >=95%?	UNKNOWN: no quality data	UNKNOWN: no quality data	UNKNOWN: no quality data

Upside incremental operating 2.8 and endpoint cash 5.9 are modeled passes of those endpoint tests only. A redesign needs its own coherent model. Do not add the independent valuation NPV to these benefits.

Valuation | One price-date, matched rates

Independent two-year lab, lakh rupees at time 0 price date. Year-end cash; real rate 10%, general inflation 5%; nominal flows at their payment-date prices; no tax or fees.

Today cost	150.00
Real annual receipt	100.00
Real annual discount rate	10.00%
General annual inflation	5.00%

Valid inputs? (1=yes)	1.0000
Exact nominal annual rate	15.50%
Year 1 nominal receipt	105.0000
Year 2 nominal receipt	110.2500

Real NPV	23.5537
Nominal NPV	23.5537
Nominal minus real residual	0.0000

Wrong: nominal flows / real rate	36.5702
Wrong: real flows / nominal rate	11.5412

NPVs match before rounding: 23.553719. Wrong mismatches are diagnostic comparisons, not alternative valid valuations. Different specific inflation, taxes or price dates requires a new model, not a blind tax shortcut.

Registry | Every numerical input is invented

Date is the model version, not a claimed company observation date. Scenario rows and opening ledger are explicitly assumed/forecast within the classroom; no real owner is asserted.

Input/source ID	Input cells / meaning	Date	Unit	Period	Fictional owner	Status	Limitations
D85/86-B	Inputs!B6:E10 revenue/cost/collection/capex	2026-11-13	lakh INR	B twelve months	Case author	Synthetic forecasts	Not causal evidence or probabilities.
D88-OPEN	Inputs!B12:E15 opening cash/PPE/AR/equity	2026-11-13	lakh INR	Start of B	Case author	Synthetic assumptions	No real accounting ledger.
D88-DEP	Inputs!B11:E11 depreciation 2 in all plans	2026-11-13	lakh INR	B twelve months	Case author	Synthetic assumption	Fixed even in capex+1 test.
D85/86-GATE	Inputs!B18:B20 floor/operating/quality	2026-11-13	lakh / ratio	B horizon	Case author	Stipulated gates	Dates and quality absent.
D88-VAL	Valuation!B5:B8 cost/cash/rates	2026-11-13	lakh / annual rates	Today + Y1/Y2	Case author	Independent synthetic lab	Not extra SME value.
ICAEW20	Input/process/output and controls	Rechecked 2026-11-13	Practice guidance	2024 edition	ICAEW	Published guidance	Self-review is limited; no independent audit implied.
ACCA-INFL	Matching nominal/real rates and flows	Rechecked 2026-11-13	Method guidance	Two-year case	ACCA	Published guidance	Historical tax example not adopted.

ICAEW: https://www.icaew.com/technical/technology/excel-community/20-principles-for-good-spreadsheet-practice-2024-edition

ACCA: https://www.accaglobal.com/gb/en/student/exam-support-resources/fundamentals-exams-study-resources/f9/technical-articles/inflation-investment-appraisal.html

Failure log | Observed authored-file tests

Native LibreOffice outcomes are written here only after the test executes. They do not certify a learner file, the truth of forecasts, compatible behavior in every spreadsheet app or independent review.

Controlled case	Expected / mechanism	Observed outcome
Baseline/scenarios	Independent statement/identity checks	PASS: independent statements, identities and matched NPVs.
Collections+1	AR-1/cash+1; NI/assets unchanged	PASS: native observed changes matched expected deltas.
Capex+1	Cash-1/PPE+1; NI unchanged	PASS: native observed changes matched expected deltas.
Variable cost+1	NI/cash/equity-1	PASS: native observed changes matched expected deltas.
Negative/text/blank	Valid flag 0, outputs unavailable	PASS: validity 0, output error; blank/text not treated as 0.
Impossible collection/dep.	Valid flag 0, outputs unavailable	PASS: validity 0, output error; blank/text not treated as 0.
Broken formula reference	Error detected, audit flag 0	PASS: reference error propagated; audit valid flag 0.
Inserted input row	Reference update preserves outputs	PASS: inserted input row preserved all referenced outputs.
Zero revenue denominator	Margin unavailable, no fake zero	PASS: zero revenue allowed; margin undefined, not 0.
Zero valuation denominator	Invalid rate flagged, NPV unavailable	PASS: rate -100% invalid; NPV unavailable.
Restore baseline	Original outputs restored	PASS: baseline/scenario/valuation outputs restored.

Keep a separate record of your own attempts, error cause, formula/input correction and changed-context retest. Do not copy this observed authored-model log as proof that your file was tested. No actual external review, protection change, credential or account action.